

Grounding the Loop on Both Sides: Interaction as a Third Test-Time Compute Axis, and Why Its Gains Are Invisible Without Grounded Evaluation

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Abstract

Test-time compute is scaled today along two axes—longer *reasoning* and more *sampling*—that share a hidden property: both are *internal*. Every additional token is generated from the same frozen weights and the same prompt, so neither axis can tell the model anything it does not already know. A third axis, *interaction*—propose an artifact, observe it with an external instrument, revise—is not subject to that internal ceiling, because each cycle imports a real observation of the artifact’s behavior.

We argue that a single variable, **grounding**, governs this third axis, and that it must hold on *both* sides of the loop: the feedback that drives revision must come from an instrument that actually observes the defect, and so must the metric that scores the result. We state the argument in one paragraph of information theory (formalized in the appendix) and then test it experimentally. On hard coding tasks at a matched token budget, reasoning-only and best-of- N scaling saturate—the latter even with an *oracle* verifier—while every interaction strategy keeps climbing, the proposer-reviewer harness reaching a strict 100% pass rate with zero seed variance, an effect that replicates across three model families and a held-out suite. On rendered visual artifacts we show that the standard vision-language-model judge is *structurally blind* to the defects the loop fixes—it rates 14 of 15 broken single-shot figures “perfect”—while a deterministic DOM-geometry instrument reveals 40–74% defect reductions across four modalities, every one statistically decisive. Swapping only the reviewer’s instrument exposes the same asymmetry on the feedback side: a VLM reviewer makes slide layout *worse*, while a geometric reviewer repairs it. Finally, the harness’s behavior partially distills into an 8B student. Interaction scaling is real, distinct from reasoning and sampling, and predictable—but its gains are visible only when both the feedback and the metric are grounded.

Code, task suites, prompts, and deterministic instruments:

<https://github.com/19PINE-AI/interaction-scaling>

Website (with an interactive single-shot-vs-reviewed results browser):

<https://01.me/research/interaction-scaling>

1 Introduction

Scaling test-time compute has become the dominant lever for improving large models on artifact-producing tasks: code, web pages, slides, figures, animations, video edits, research

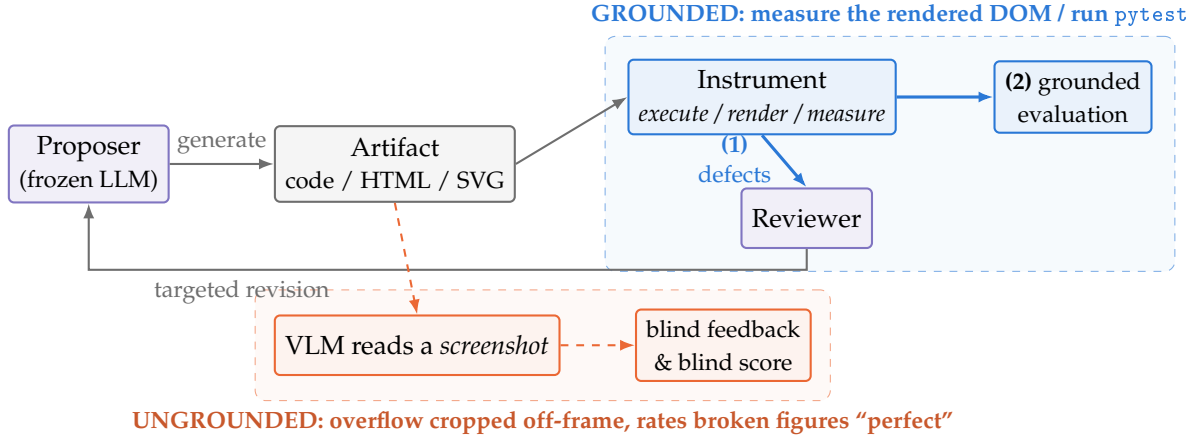


Figure 1. Grounding must hold on both sides of the interaction loop. A proposer emits an artifact, and a grounded instrument *executes, renders, or measures* it. That single observation does double duty. (1) As grounded *feedback*, it becomes a defect list the reviewer turns into a targeted revision—the channel that lets interaction import information from outside the weights and escape the internal ceiling (Section 2). (2) As grounded *evaluation*, it is the score that makes the gain measurable. Substituting the default VLM-on-a-screenshot judge (orange lane) breaks *both*: a screenshot drops off-canvas overflow and fine overlap before the model ever sees them, so it neither scores the defect nor—as a reviewer—fixes it.

reports. The literature has formalized two axes. *Reasoning scaling* [4, 19, 25, 26] spends more tokens thinking before committing to an answer. *Sampling scaling* [1, 24] draws more independent attempts and selects one. The two look different, but they share a property that this paper treats as fundamental: both are **internal**. Every additional token, whether spent on a longer chain of thought or another sample, is generated from the same frozen weights and the same fixed prompt. More internal compute reorganizes information the system already has; it imports none.

A third axis breaks this closure: *interaction* with an external instrument that observes the artifact itself—run the tests, render the page and measure the layout, issue the search query. Such an observation is not a function of the weights. It reports how the artifact actually behaves, and it can carry information the model never had. This axis is well established empirically [6, 17, 18, 27], but accounts of *why* and *when* it should beat internal scaling remain informal and, we argue, half-stated. Practitioner reports at industrial scale sharpen the stakes: in a controlled 900-run deployment study at ByteDance, frontier coding models pass the functional-correctness bar yet miss on delivery quality until a feedback “harness” is added—and that report had to invent a multi-axis quality metric before the gap was even visible [7]. That is a field observation of exactly the two-sided problem we formalize below (detailed in Section 9).

Grounding is the variable, on both sides of the loop. Our thesis is that one variable governs interaction scaling: **grounding**—the property that a feedback signal is produced by an instrument observing the artifact’s actual form or behavior, rather than by a model rendering an opinion about it. And grounding must hold on *both* sides of the proposer–reviewer loop (Figure 1):

1. **Grounded feedback** (the signal that drives revision): run the code and read the failing test, render the HTML and measure its bounding boxes, issue the query and read the result. This is the side prior work emphasizes, and a one-paragraph information argument

(Section 2) says it is exactly what lets interaction escape the ceiling that bounds reasoning and sampling.

2. **Grounded evaluation** (the metric that *measures* the gain): the same observation, used as the score. This side is almost always overlooked, and getting it wrong silently hides the effect. The second point is not a detail. The default evaluator for visual artifacts is a vision–language model (VLM) reading a screenshot, and a screenshot is an instrument whose channel *drops the defect before the judge ever sees it*: content overflowing the canvas is cropped off-frame, and small overlaps fall below the model’s visual acuity. On dense academic figures this judge rates 14 of 15 single-shot renders “perfect” when a deterministic measurement finds only 3 actually clean (Section 6). Worse, when the same VLM is used as the *reviewer*, its edits make slide layouts worse (Section 5). Ungrounded on either side, the third axis either does not fire or cannot be seen.

Contributions.

1. **A grounding framework (Section 2).** A two-category feedback taxonomy—*ungrounded* model opinion vs. *grounded* instrument observation—together with a *coverage* principle: grounded feedback helps exactly as far as the instrument’s observational reach. A one-paragraph information argument (formal statement in Appendix A) explains why internal scaling saturates, and a symmetric observation says an ungrounded *metric* cannot measure the gain. Beyond the documented *biases* of model-as-judge evaluation [28], we isolate a sharper failure: for artifacts whose quality is a measurable physical property, a screenshot judge is *structurally blind*—not merely noisy—which silently hides a real interaction-scaling effect.
2. **Interaction keeps scaling where internal compute stops (Section 4).** At a matched token budget, reasoning-only and best-of- N saturate—the latter even with an oracle verifier—while every interaction strategy approaches a perfect pass rate, the proposer–reviewer harness doing so at the lowest token cost with zero seed variance.
3. **The feedback instrument must observe the defect (Section 5).** Holding the task suite fixed and swapping only the reviewer’s instrument: execution feedback converges far cheaper than critique on behavioral bugs; a linter (grounded, but observing only form) buys nothing on them; and a screenshot-reading VLM reviewer makes slide geometry *worse* while a geometric reviewer repairs it.
4. **So must the metric (Section 6).** The standard VLM judge passes 14 of 15 broken figures; a deterministic DOM-geometry instrument instead reveals large, statistically decisive defect reductions under grounded feedback across four visual modalities measured identically—gains the standard metric cannot even see.
5. **Robustness and internalization (Sections 7 and 8).** The effects replicate across three model families (including the geometry result with Gemini 3.1 Pro as proposer), on a held-out suite, and across a budget-allocation sweep. The harness’s behavior also partially distills into an 8B student at $\sim 10\times$ lower deployment cost.

2 Two Internal Axes, One External Axis

Our contribution is not a new self-correction method but an account of *why* interaction is a separate scaling axis, *when* a feedback channel helps (a coverage principle over a simple taxonomy), and a previously-missing requirement: that the *evaluation* be grounded too, without which the effect is unmeasurable.

2.1 Internal vs. external test-time compute

Reasoning and sampling differ in mechanics but not in kind. A longer chain of thought re-derives consequences of what the weights and prompt already contain; best-of- N re-draws from the same distribution and selects. In both, every token is manufactured from the same two sources—frozen weights, fixed prompt—so we call them **internal** scaling. **External** scaling is different in kind: the artifact is handed to an instrument outside the model—a test runner, a layout engine, a search index—and the instrument’s observation is fed back into generation. The observation is manufactured from the *artifact’s actual behavior*, not from the model’s beliefs about it.

2.2 A feedback taxonomy, and the coverage principle

We classify a reviewer’s feedback channel by *who produces the signal* (Figure 2):

- **Ungrounded feedback** is a model’s opinion. An LLM re-reads its own artifact and critiques it; a VLM looks at a screenshot and rates it. No instrument touches the artifact.
- **Grounded feedback** is an instrument’s observation. Its subtypes are named by *what the instrument observes*: **static form** (linters and type checkers inspect the source), **execution behavior** (tests run it and report the failing assertion), **rendered geometry** (a layout engine reports every element’s true bounding box), **temporal behavior** (the same measurement per animation frame), and **external facts** (a search engine checks a claim against the world).

Grounding alone, however, is not sufficient, and this is the second half of the taxonomy:

The coverage principle. A feedback channel helps exactly as far as its instrument’s observational reach. A linter is grounded but observes only form, so it cannot see a runtime bug; a screenshot is an observation whose channel crops the defect out, so a VLM judging it cannot see broken geometry. The instrument must *observe the property that is broken*.

The coverage principle is what unifies the paper’s results: it predicts that execution feedback beats critique on behavioral bugs while a linter does not (Section 5), that a screenshot-fed VLM fails both as a reviewer (Section 5) and as a judge (Section 6), and that a deterministic geometry instrument succeeds at both.

2.3 Why internal scaling saturates

The reason is one sentence long: *a model re-reading its own output cannot learn anything it did not already know*. A critique produced by the same weights that produced the artifact is post-processing, and post-processing cannot add information about the correct answer beyond what the weights and prompt already carried. Sampling is bounded for a different but equally

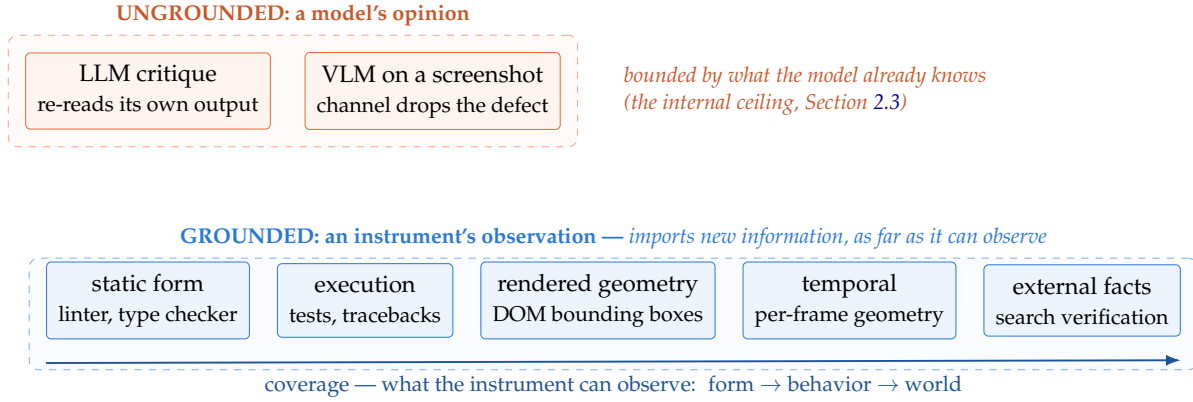


Figure 2. The feedback taxonomy. Feedback is either a model’s opinion (*ungrounded*, left) or an instrument’s observation of the artifact (*grounded*, right); grounded subtypes are named by what the instrument observes, and the *coverage* axis orders them by reach. Ungrounded feedback is subject to the internal ceiling (Section 2.3); grounded feedback escapes it, but only for defects inside the instrument’s coverage—a linter cannot see a runtime bug, and a screenshot cannot see off-canvas overflow.

simple reason: best-of- N can only *select* among the candidates the model actually draws, so even a perfect verifier cannot return a program the model was never going to write. Longer thinking rearranges the known; more samples explore the already-likely. We call the resulting plateau the **internal ceiling**. The formal version is a three-line data-processing-inequality argument, and we defer it to Appendix A; the inequality bounds an information channel, not achievable quality directly, so throughout the paper the evidence is experimental. Section 4 measures the ceiling directly: on hard code, reasoning-only and *oracle-verified* best-of- N both flatten with budget while interaction strategies climb past them to a perfect pass rate (Figure 3).

Grounded feedback escapes the ceiling because the instrument’s observation is computed from the artifact’s real behavior—a failing assertion, a measured bounding box, a search verdict—and conditioning the *next* proposal on that observation lets the model synthesize a candidate it would never have sampled unaided. This is the crucial difference from selection (which cannot create a missing candidate) and from reasoning (which adds no information): interaction is a new information channel, not a constant-factor speedup.

The symmetric claim: an ungrounded metric cannot measure the gain. The same argument applies to the *scorer*. A judge that is itself a model looking at a lossy view of the artifact—a VLM reading a cropped screenshot—cannot certify an improvement that lives in the part of the artifact its channel drops. Grounding the evaluation is therefore not optional polish: it is a precondition for *detecting* interaction scaling on any modality whose quality is not fully visible to the default judge. Section 6 shows this is exactly the situation for rendered visual artifacts.

2.4 Three testable predictions

(P1) At a matched token budget, reasoning-only and best-of- N scaling saturate strictly below what interaction reaches (Section 4). **(P2)** Holding the task suite fixed and swapping only the reviewer’s feedback channel, improvement tracks the instrument’s *coverage* of the defects present—not the act of reviewing (Section 5). **(P3)** On modalities whose quality is invisible to the default judge, the gain is only detectable with a grounded metric, and an ungrounded

Table 1. Modality instantiations. Each modality pairs a grounded feedback channel (named per the taxonomy of Figure 2) with the instrument that both drives revision and scores the result. “†” marks the four modalities scored by the deterministic DOM-geometry instrument and analyzed jointly in Section 6.

Modality	Feedback channel	Grounded instrument	Hard suite (N)
Code	execution	pytest pass/fail + traceback	15 (+11 deep-spec)
Academic figures [†]	geometry	DOM geometry + alignment	20
Slides [†]	geometry	DOM geometry + alignment	12 real-paper
Web pages [†]	geometry	DOM geometry, multi-width	20
Animations [†]	temporal	DOM geometry, per-frame	20 SVG/CSS
Video editing	exec. + temporal	script exec + Gemini-native rubric	15
Deep research	external facts	search-grounded factual rubric	15

reviewer can even regress quality (Sections 5 and 6).

3 Setup: A Budget-Aware Proposer–Reviewer Harness

Architecture. The harness is pure scaffolding around a *frozen* frontier model (Claude Sonnet 4, `claude-sonnet-4-20250514`, temperature 0 unless noted): no fine-tuning, no retrieval, no learned controller. A **proposer** emits an artifact. An **instrument** executes, renders, or measures it to produce a grounded signal. A **reviewer**—the same model in a diagnostic role—converts that signal into a typed defect list, and the proposer revises. The loop repeats up to an iteration cap and *keeps the best-scoring iteration*, so the reviewed condition can never score below single-shot under the same metric. Single-shot is one proposer call.

The three grounded instruments. Every modality is paired with a deterministic or near-deterministic instrument that both *feeds back* and *scores*:

- **Execution** (pytest): exact pass/fail plus the failing assertion and traceback. Each task ships a multi-assertion suite targeting the edge cases that distinguish correct from plausible-but-wrong (e.g. the semantic-version comparator is checked on pre-release precedence, not just major/minor/patch), and the deep-spec suite is additionally validated against a reference implementation. A “pass” therefore certifies the specified behavior, not mere lint-cleanliness—and the oracle best-of- N baseline of Section 4 selects against these *same* tests, so its ceiling is measured by an identical, strong criterion.
- **Rendered geometry**: the artifact is rendered headless and every element’s true bounding box is read from the layout engine. We compute, exactly: text-on-text overlap (≥ 6 px), out-of-bounds/clipping, container overflow, document overflow (> 16 px), and *box-group misalignment* (rows or columns of card-like boxes flagged for unequal size, misaligned far edges, or uneven gutters). Web pages are scored at desktop and mobile widths; animations are probed at each sampled frame.
- **Native video** (Gemini 3.1 Pro): the rendered clip is judged whole against per-requirement binary checks, a near-grounded substitute where pixels, not a still, are observed.

Where no deterministic instrument exists (flowing web content, factual research), we fall back to a binary per-requirement rubric, flagged as such.

All artifacts are produced under a common *design-principle* generation prompt (proximity, alignment, repetition, contrast, deliberate color) so that single-shot quality is already strong. The headroom we measure is therefore a property of genuine task difficulty, not of a weak prompt.

Budget and protocol. A run has a token budget B split across proposer and reviewer calls; Section 7 sweeps the split, and Table 1 lists the modality instantiations. Unless noted, the proposer runs at temperature 0 and each modality is evaluated over three independent on-policy seeds, keeping the best-scoring iteration (cap ≤ 3 for geometry, ≤ 5 for code). The unit of comparison is the (task, seed) pair. For each result we report a two-sided paired sign test over decisive pairs and, for the geometric and code effects, a paired-bootstrap 95% confidence interval on the effect size; exact values accompany each figure.

4 External Scaling Keeps Going Where Internal Scaling Stops

The saturation experiment (P1). On the hard code suite at a fixed per-task token budget we compare four strategies: reasoning-only (extended thinking), best-of- N sampling with an *oracle* verifier, a single-agent loop, and the proposer-reviewer harness. Figure 3 shows the result, which is the experimental content of the internal ceiling (Section 2.3): both internal strategies flatten as the budget grows, while *every* feedback-iterating strategy climbs past them toward a perfect pass rate. The best-of- N comparison is the sharpest, because its verifier is the ground-truth test suite itself—its ceiling is effectively pass@ N . The tasks it never solves are tasks where no passing sample exists among anything the model draws within the budget; a perfect selector cannot return a candidate the proposer never writes. The harness solves those same tasks by conditioning the next draw on execution feedback. The separation is information-theoretic, not a tuning artifact.

Code as the clean case study. Code is the cleanest demonstration because the instrument is exact and the score is objective. The harness recovers every first-shot failure with zero regressions on both the development suite and a harder deep-spec suite of from-scratch implementations (a JSON parser, a spreadsheet-reference resolver, a minimal edit-script diff, an expression evaluator, each validated against a reference solution), lifting both suites to a strict 100% pass rate (Figure 4). The mechanism is consistent across recovered traces. Turn 1 emits code that compiles but mishandles an edge case—off-by-one boundaries, semantic-version comparison, CIDR arithmetic. Turn 2 receives the failing assertion, the offending input, and the expected-vs-actual values, and rewrites the localized block. Turn 3 confirms. Reasoning alone cannot supply this: the model has no way to know its boundary condition is wrong until a concrete test exercises it.

Architecture buys efficiency and reliability, not ceiling. Among interaction strategies the ceiling pass rate ties within seed noise; what separates them is cost and variance (Figure 5). The proposer-reviewer harness is the most token-efficient and the only variant that reaches 100% in *every* seed. Context isolation (the reviewer sees only what it needs to localize defects), structured critique (a typed defect list, not raw stderr), and role specialization explain the efficiency.

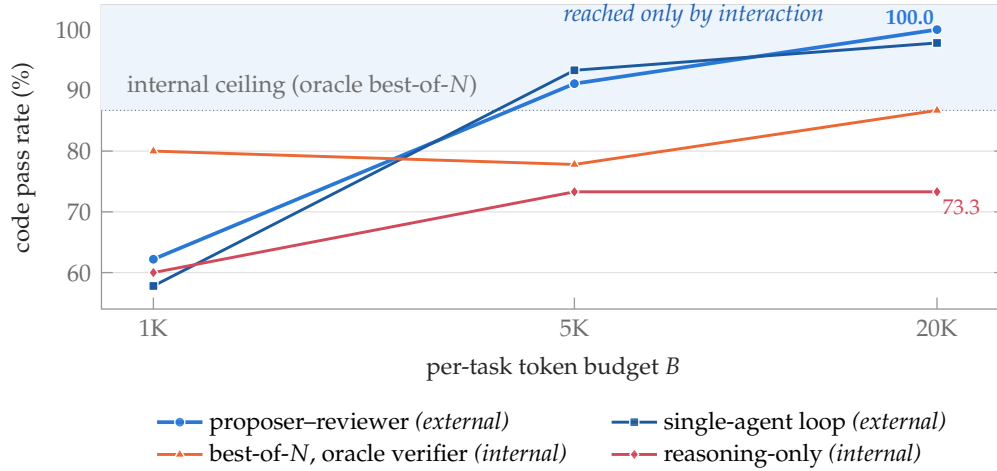


Figure 3. Internal scaling saturates; external scaling does not (P1). Code pass rate vs. per-task token budget on the 15 hard tasks (3-seed means; full table with SDs in Table 2). The two internal strategies flatten with budget: reasoning-only at its information ceiling, and best-of- N at 86.7% *even with an oracle* (ground-truth-test) verifier—the high-probability support of the model’s own output distribution. The two external strategies import execution feedback each cycle and climb past the oracle-sampling ceiling, the proposer-reviewer harness to a strict 100% with zero seed variance. This is the internal-ceiling prediction of Section 2.3, made concrete.

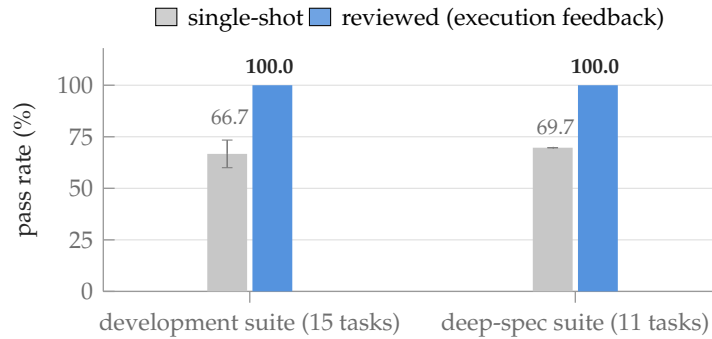


Figure 4. Execution feedback recovers every first-shot failure on both code suites. Single-shot vs. harness-reviewed pass rate (development suite: 3-seed mean, whisker ± 1 SD; deep-spec: sign test $p = 0.002$). Every failing run is recovered and no passing run regresses; the reviewed bars carry zero seed variance.

Built-in early stopping. The loop does not over-revise: on code, *every* already-passing single-shot task is submitted at iteration 1 with no revision and no wasted tokens. The grounded signal—a failing test—is the trigger; without it the loop is silent. This is why the harness’s token overhead over single-shot stays modest despite a generous iteration cap.

5 The Feedback Side: The Instrument Must Observe the Defect

The saturation experiment compared strategies. This section holds everything else fixed—task suite, proposer model, number of reviewing passes—and swaps *only the reviewer’s feedback channel*, testing the coverage principle (P2) directly on both a textual and a visual modality.

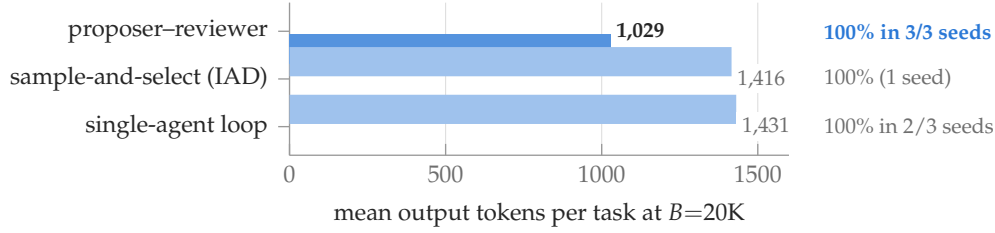


Figure 5. Same ceiling, different cost and reliability. All three interaction strategies use execution feedback and tie on pass rate within seed noise (sign test $p > 0.6$); the proposer-reviewer harness converges with $\sim 28\%$ fewer tokens and is the only variant at 100% across all seeds (Table 5).

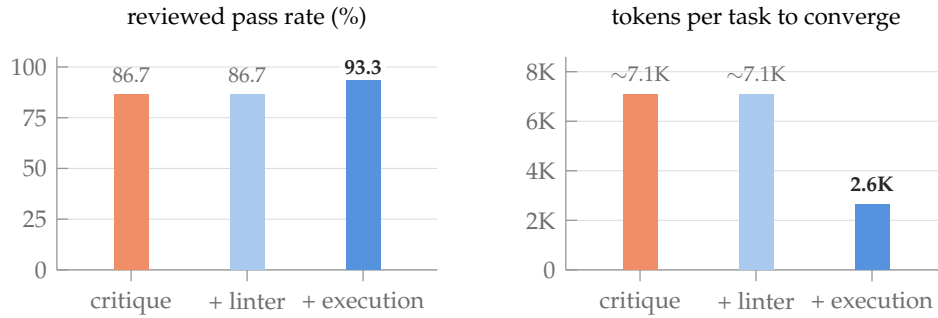


Figure 6. Swapping only the feedback signal on a fixed code suite: improvement tracks coverage, and cost tracks it decisively. One reviewing pass per arm. Ungrounded critique (orange), critique + linter (light blue: grounded, but observing only *form*), and critique + test execution (blue: grounded, observing *behavior*). The linter arm matches bare critique on these runtime-logic bugs—exactly what the coverage principle predicts—while the execution arm reaches a higher ceiling at $\sim 2.5\times$ lower cost and fewer iterations (1.53 vs. 2.07; per-seed detail in Table 4).

On code: execution feedback wins where its coverage lies. We run three arms on the fixed code suite, each with *exactly one* reviewing pass: an ungrounded critique reviewer (the LLM re-reads the code), a static-form reviewer (the same critique plus ruff linter output), and an execution reviewer (the same critique plus the failing test’s traceback). Any separation therefore isolates the *signal*, not the act of reviewing. Figure 6 shows both outcomes. The pass-rate separation is modest and we read it cautiously, since it rests on a single leaner-budget seed. The unambiguous separation is *cost*: the execution arm converges roughly $2.5\times$ cheaper and in fewer iterations, because a concrete failing assertion localizes the fix where an ungrounded critique gropes. The static-form arm is the coverage principle’s cleanest prediction: the linter is a genuinely grounded instrument, but the seeded bugs are runtime-logic errors that leave no trace in form, so it buys nothing over bare critique—grounding helps only as far as the instrument can see.¹

On slides and figures: an uncovered instrument makes things worse. The decisive form of the control is visual. We fix the task suite and the proposer, and give the reviewer either (i) the deterministic geometry report—measured boxes, exact overlaps—or (ii) a VLM’s reading of a screenshot, an instrument whose channel crops the very defects at issue. Because the two arms

¹Symmetrically, the principle predicts a static-form lift on a bug set that *is* visible in form (type confusions catchable by `mypy`, taint patterns catchable by `semgrep`). Constructing that suite is future work; on our behavioral bug set the null result is the prediction.

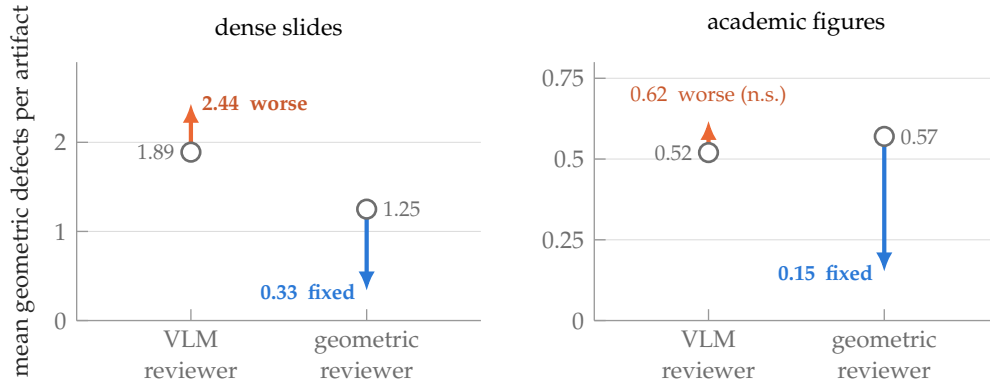


Figure 7. Same tasks, same proposer, one reviewing pass—the arrow’s direction is set by the reviewer’s instrument. Each arrow runs from that arm’s own single-shot baseline (open circle; the arms are separate runs, hence differing baselines) to its reviewed result, in mean geometric defects per artifact (lower is better; note the differing y scales). The screenshot-fed VLM reviewer moves *up* on slides and directionally *up* on figures; the deterministic geometric reviewer moves sharply *down* on both (slides -73% , $p = 0.0018$; figures -74% , $p = 7 \times 10^{-4}$).

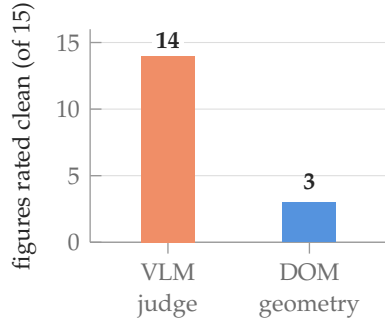
are separate runs, each is compared against its *own* single-shot baseline, and the load-bearing contrast is the *direction* each reviewer moves that baseline (Figure 7). The geometric reviewer sharply reduces real layout defects on both modalities. The VLM reviewer does not: it *increases* defects on slides and is directionally worse on figures. Blind to the true geometry, its edits break alignment about as often as they fix it—one reviewing pass, opposite sign, set entirely by whether the signal’s coverage includes the defect. (On the same slides the binary VLM *rubric* simultaneously saturates, which is the evaluation-side half of the story; Section 6 takes it up.)

What this establishes. On the feedback side, grounding is necessary but the operative quantity is *coverage*: a grounded instrument that cannot observe the defect class (the linter, on runtime bugs) matches ungrounded critique, and an instrument whose channel actively drops the defect (the screenshot) is worse than no reviewer at all. The remaining question is symmetric: what happens when the *metric* is such an instrument.

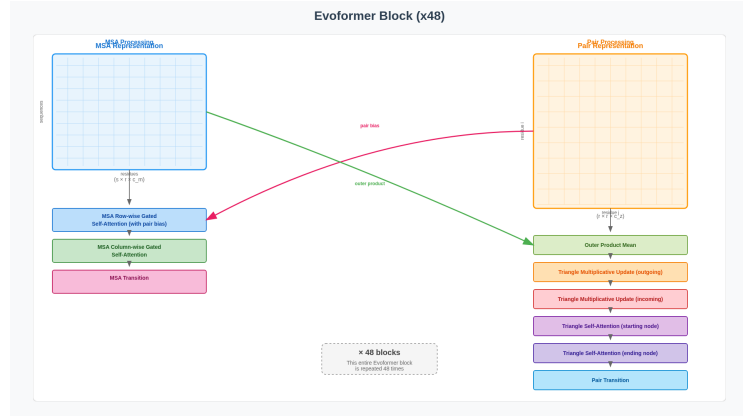
6 The Evaluation Side: The Standard Metric Cannot See Any of This

The visual modalities expose the symmetric half of the framework (P3): even when grounded feedback *does* fix the artifact, an ungrounded metric cannot see the fix.

The default VLM judge is structurally blind. The field’s default evaluator for visual artifacts is a binary per-requirement rubric scored by a VLM over a screenshot. On an initial probe of 15 dense academic-paper architecture figures—even with native-resolution quadrant tiling—this judge rates 14 of 15 single-shot renders “perfect,” i.e. satisfying every *content* requirement it checks: the right components present, the labels correct. By direct inspection most of those renders are broken, with superimposed section titles and labels spilling out of boxes (Figure 8b). The failure is mechanical, not fixable by prompting the judge better. Screenshots are captured at a fixed resolution, so content overflowing the canvas is cropped *off-frame before the image reaches the judge*—the defect is literally absent from the pixels the model sees—and small

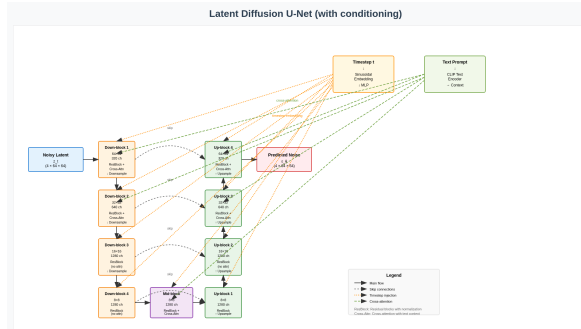


(a) The VLM passes 14/15; the deterministic instrument, 3/15.

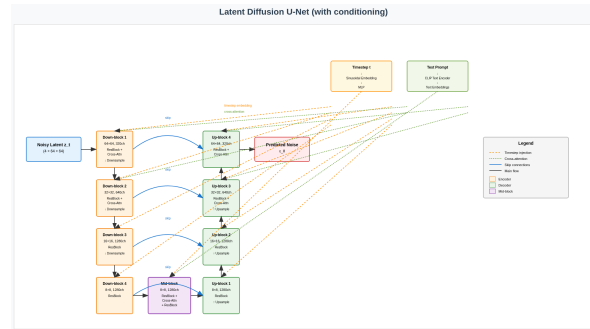


(b) A single-shot figure the VLM rated “perfect”: both section titles are superimposed (“MSA Processing” over “MSA Representation”; “Pair Processing” over “Pair Representation”) and axis labels collide with the matrix borders.

Figure 8. The default VLM judge is structurally blind to layout defects. (a) On the same 15 single-shot academic figures, the VLM-on-a-screenshot judge rates 14 “perfect” while the DOM-geometry instrument finds only 3 actually clean; the other 11 are broken in ways cropped off-frame or below VLM acuity. (b) A representative “perfect”-rated render whose text-on-text overlaps are exactly what the geometry check flags and the screenshot judge misses.



(a) **Single-shot.** Conditioning blocks crowd the down-block column, edges route through boxes and tangle at the center, and the legend collides with the decoder—all defects the geometry instrument flags and the VLM cannot see.



(b) **After grounded feedback.** The same task: conditioning blocks lifted clear, columns spaced and aligned, skip connections deconflicted, legend boxed in free space.

Figure 9. What grounded geometric feedback actually fixes. The same academic-figure task, single-shot (a) vs. after the propose→measure→revise loop (b). This is the per-artifact reality behind the aggregate reductions of Figure 10, and exactly the difference a screenshot judge is blind to.

in-frame overlaps fall below its acuity. The deterministic DOM-geometry instrument, reading the same artifacts’ true bounding boxes, finds only 3 of 15 actually clean (Figure 8a). The gap is not a wrong answer to the same question; it is the judge being *structurally unable to observe* the property that matters—an instrument whose coverage excludes the defect, deployed as the score. (This probe is a deliberately hard, unconstrained set; the suites scored below are generated under the design-principle prompt of Section 3 and are correspondingly cleaner single-shot.)

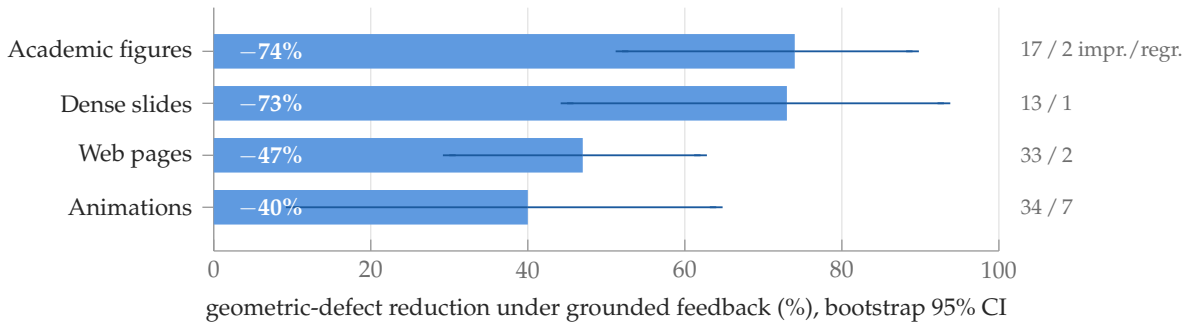


Figure 10. Grounded geometric feedback removes real layout defects on all four visual modalities. Mean defect reduction under one identical configuration (3 seeds; whiskers are paired-bootstrap 95% CIs, all excluding zero; right column counts improved vs. regressed task-runs; all paired sign tests $p < 2 \times 10^{-3}$; full statistics in Table 6). The standard VLM-on-a-screenshot metric measures none of this.

A grounded instrument reveals a large, decisive effect on all four modalities. We score four visual modalities with the same DOM-geometry-plus-alignment instrument under one identical configuration (Sonnet 4, temperature 0, design-principle prompt, three seeds, propose→measure→feed-exact-defects-back→revise, ≤ 3 iterations). Figure 10 summarizes and Table 6 in the appendix gives the full statistics: grounded geometric feedback removes a large fraction of real layout defects on *all four* modalities, with every bootstrap confidence interval excluding zero and improvements outnumbering regressions by roughly an order of magnitude. The magnitudes track single-shot headroom—dense responsive web pages are defect-rich out of the box, while a frontier model under a design-principle prompt already lays out many figures and slides cleanly. Animations carry the widest interval because their per-task defect counts are heavy-tailed, so we read the sign test as the robust statement there and the mean as indicative.

Is the reduction circular? The instrument both *drives* the revision and *scores* it, and the harness keeps the best-scoring iteration, so *some* reduction is mechanically guaranteed. The honest questions are how large it is and whether it reflects quality a human would care about. Three facts argue it is not merely an artifact of optimizing the reported number. **(i) The defects are real:** single-shot renders carry them at high rate (Figure 8b), and the thresholds are deliberately conservative, so a flagged defect is visible, not sub-perceptual. **(ii) The magnitude is not preordained:** keeping the best of three iterations could in principle move the metric by almost nothing; that grounded revision instead removes most of the real defects (Figure 10) is a property of the feedback, not of the keep-best rule. **(iii) The ungrounded-reviewer ablation is the decisive control:** scored by the *same* metric, the VLM-feedback arm fails to reduce it and worsens slides (Figure 7). So the reduction tracks the grounding of the feedback, not the act of optimizing the scorer—a conclusion the model-free cross-model replication (Section 7) reinforces. What remains open, and is the principal caveat of the geometric measurement, is a quantitative human-preference study confirming that DOM-defect reduction maps onto perceived-quality gains across the full suite.

Two modalities where the honest answer is “no headroom.” *Video editing*, scored by Gemini 3.1 Pro’s native full-clip rubric (a near-grounded instrument that observes pixels, not a still), is already strong single-shot, so the reviewed lift is small and not significant; a

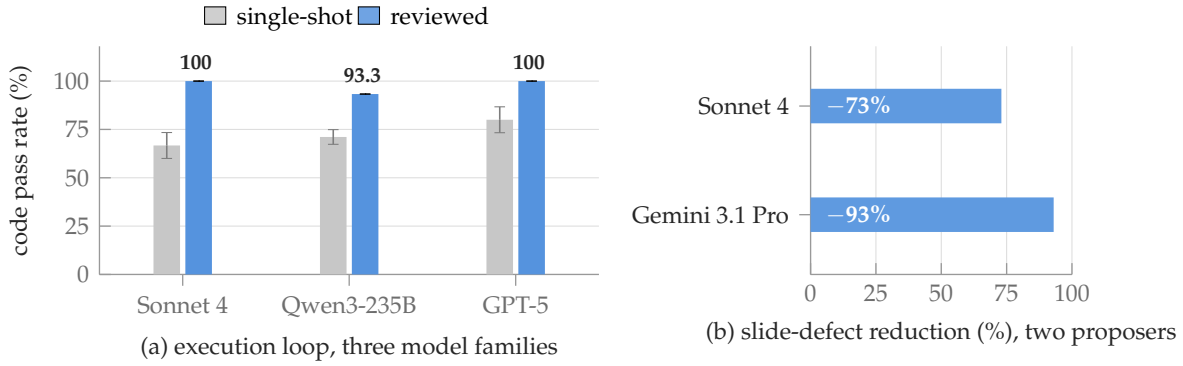


Figure 11. Both grounded loops replicate across model families. (a) Single-shot vs. harness-reviewed code pass rate for three families, three seeds each (lifts +33.3/+22.2/+20.0 pp; whiskers ± 1 SD across seeds). The reviewed bars carry *zero* seed variance for all three families. **(b)** The deterministic geometric loop on dense real-paper slides, scored by the same model-free DOM instrument, reproduces with Gemini 3.1 Pro as proposer and is larger there (19 of 20 decisive task-runs improve): Gemini starts more defect-prone and the loop removes almost all of the excess.

hardened multi-step suite de-saturates it. *Deep research*, scored against exact provided facts, saturates: a frontier model knows well-documented facts and does not take the planted traps, so the factual-feedback lift is bounded by a near-zero single-shot error rate. This is a genuine limit of the factual modality, since de-saturating it would require facts the judge itself cannot grade. We report both as scoped negatives rather than headline lifts (Table 17).

7 Robustness

Cross-model, code. The effect is architectural, not a Claude artifact. On the hard code suite, over three independent on-policy seeds per family, the harness lifts all three families we test—Sonnet 4, Qwen3-235B, and GPT-5, the last from a higher single-shot baseline (Figure 11a). The signature detail is the variance: single-shot fluctuates seed-to-seed for every family, while the reviewed ceiling has *zero* seed variance for all three. The grounded loop converges to the same ceiling regardless of the starting draw.

Cross-model, grounded geometry. To rule out a Claude-specific or prompt-specific explanation of the geometric effect, we run the same deterministic geometric-feedback harness with **Gemini 3.1 Pro** as the proposer on the dense real-paper slide suite, scored by the identical model-free DOM instrument. The effect reproduces and is in fact larger (Figure 11b): Gemini’s single-shot slides are more defect-prone than Sonnet’s, and the grounded loop removes almost all of the excess. Since the instrument depends on no model, the effect is neither proposer- nor scorer-specific.

Held-out generalization. On a 32-task held-out code suite constructed after the method was fixed, the harness again recovers every first-shot failure with zero regressions and reaches a perfect pass rate, with no tuning to the original tasks (Table 8). The absolute lift is smaller only because the held-out single-shot baseline is higher, leaving less headroom.

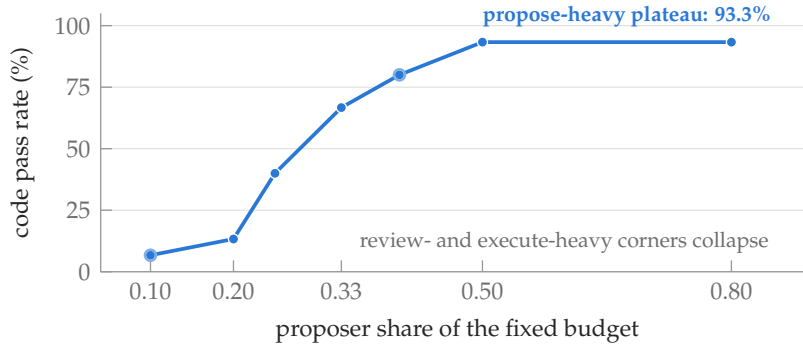


Figure 12. Pass rate is monotone in the proposer’s budget share. Nine allocations of a fixed 10K-token budget across proposer / execution / reviewer, on the hard code suite (open markers: second allocation with the same proposer share but a different execution/review split—the proposer share alone predicts the outcome). The spread across the sweep is 86.6 pp; full simplex in Table 9.

Budget allocation. Sweeping how a fixed token budget is split between proposer, execution, and reviewer produces an enormous pass-rate spread (Figure 12). The pattern is monotone in the proposer’s share: review-heavy corners collapse almost to zero, while propose-heavy allocations plateau at the top. The optimum is *propose-heavy*—allocate the majority to generation, reserving just enough for the reviewer to localize defects—and extra inference compute is better spent on additional samples than on deeper iteration past the early-stop point.

8 Internalizing the Harness into a Small Student

Two things could carry the harness’s gains: the grounded scaffolding, or behavior absorbed into the weights. This section asks how much is the latter—whether the loop’s interaction quality can be *internalized* so a small model emits high-quality artifacts in fewer passes. It is a complement to the thesis, not a substitute: the cheaper internalized proposer is still meant to run *inside* the same grounded harness, so internalization and grounding compose. We distill judge-filtered teacher trajectories from the harness into an 8B Qwen3-VL student via supervised fine-tuning.

Result. The student recovers about half the teacher’s single-sample capability on an out-of-distribution suite, and a second sample closes much of the remaining gap (Figure 13)—at roughly 10× lower deployment cost than running the harness over a frontier model. On a harder held-out suite of the teacher’s own pre-curated failures it passes a substantial fraction (Table 11), and run back inside the harness it recovers still more.

Variance is a budget (a cautionary finding). Reinforcement fine-tuning (RFT) on top of SFT improves every per-turn consistency metric we measure *and* regresses pass@k (Figure 14). The two students start nearly tied on a single sample; as samples are added, the SFT student keeps converting draws into new solved tasks while the RFT student’s curve goes flat. The reading is that for sampling-based deployment, output-distribution variance *is* the inference-scaling lever, and any post-training that reduces it spends from the same budget best-of- N draws on. Variance is not noise to minimize; it is the resource sampling converts into quality. The

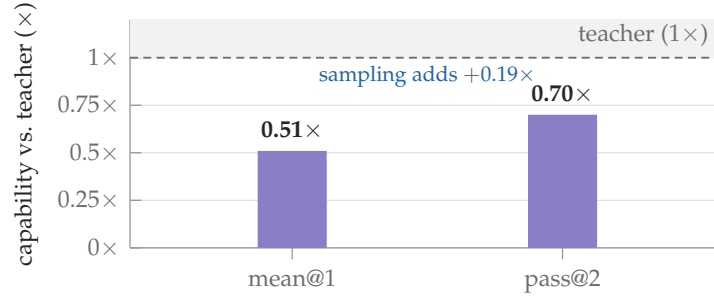


Figure 13. An 8B student internalizes much of the teacher’s interaction quality. Distilled student as a fraction of the frontier teacher on the 44-task out-of-distribution suite (Table 12): one sample recovers $0.51\times$ and two samples $0.70\times$ of teacher capability, at $\sim 10\times$ lower deployment cost. On the 18-task hard held-out suite the same student passes 44% at pass@1 and 56% at pass@3 (Table 11).

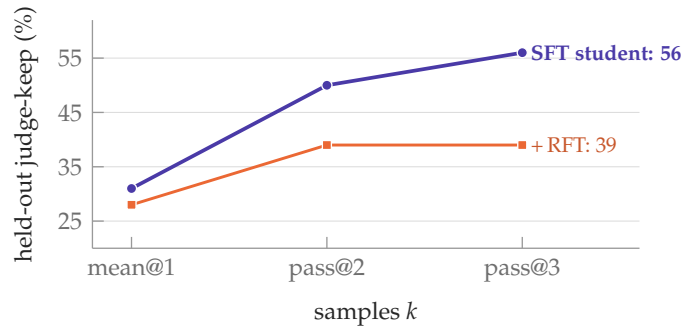


Figure 14. RFT polishes consistency but spends the variance that sampling needs. Judge-keep vs. number of samples on the 18-task hard held-out suite for the chosen SFT student and the same student after RFT. RFT improves every per-turn consistency metric (Table 15) yet regresses pass@3 by 17 pp: the near-tie at one sample and the flat curve past it show the lost quality is exactly the distributional spread best-of- N converts into solved tasks.

operational recipe is therefore: SFT to internalize interaction quality, keep the proposer’s sampling temperature, and run it inside the grounded harness.

9 Related Work

Test-time compute scaling. Two axes are established. *Reasoning* scaling (chain-of-thought, tree search, and extended thinking [4, 19, 21, 25, 26]) spends more tokens before committing. *Sampling* scaling (self-consistency and best-of- N [1, 24]) draws multiple attempts and selects. Both transform the same frozen weights and prompt—both are *internal* in the sense of Section 2, which makes precise how they are bounded—and we position *interaction* as a third, external axis that imports information the model does not have.

Self-correction and critique. Iterative refinement with model-generated feedback (Reflexion [18], ReAct [27], Self-Refine [11], CRITIC [6]) is widely used, but its reliability is contested: Huang et al. [8] show models often cannot self-correct reasoning without an external signal, and Kamoi et al. [9] chart when correction does and does not help. Our taxonomy makes the distinction explicit—*ungrounded* critique is subject to the internal ceiling, whereas refinement

against an executed or measured observation is not—and our controls (Section 5) show it is the instrument’s coverage of the defect, not the extra critique pass, that carries the gain.

Grounded and execution feedback. Tool use and execution feedback (Toolformer [16], Gorilla [12], Voyager [23], RLEF [5], and, for website generation specifically, WebGen-Agent’s multi-level visual/functional feedback [10]), together with the thinking-vs-doing analysis of Shen et al. [17], establish that acting on an environment helps. We contribute the information account of *why*, a coverage principle that predicts *when*, and, crucially, the requirement that the *evaluation* be grounded too. We also compare harness architectures (single-agent loops [22], sample-and-select [14], proposer–reviewer) under a matched budget.

Model-as-judge reliability. LLM- and VLM-as-judge evaluation is now standard [28], and its failure modes (position, verbosity, and self-preference biases, and broader reward-model fragility [2]) are documented. Our finding is sharper and, to our knowledge, not previously isolated: for artifacts whose quality is a measurable *physical* property (the geometry of a rendered layout), a screenshot judge is not merely biased but *structurally blind*—the defect is dropped by its observation channel before judgment begins—so it both fails to score the defect and, used as a reviewer, fails to fix it. This is why a deterministic instrument is necessary, not merely preferable, to detect interaction scaling on visual modalities.

Practitioner evidence at scale. Industrial deployment independently corroborates the gap our framework targets. Hong [7] reports a controlled study at ByteDance (3 frontier coding models $\times$ 3 agent frameworks $\times$ 100 runs, 900 total) on a single real product feature under a fixed prompt. Functional *correctness* exceeded 80% for every model–framework pair, yet “deliverability” axes (UI usability, reliability, maintainability, performance, compatibility) collapsed to a failing 40–60% with high run-to-run randomness. Only after adding harness “infrastructure” (context engineering, architectural constraints, and persisted team memory) did deliverability rise to $\sim$ 80%, while correctness barely moved. The same report notes that despite $>90\%$ of one team’s code being model-written, throughput rose only $1.6\times$, far short of the raw generation speedup—the macroscopic shadow of the quality single-shot generation does not deliver. This is the signature we formalize and measure: internal scaling saturates the easy-to-see axis, while the interaction loop carries the rest. Tellingly, the report had to define a multi-axis quality metric before the gap was visible at all, an industrial analogue of our grounded-evaluation requirement; its grounded fixes (e.g. browser-driven validation before commit) are instrument observations in our taxonomy, not ungrounded self-review.

Information theory. The data-processing inequality [3] underlies the formal version of the internal ceiling (Appendix A): any post-processing of the model’s outputs (reasoning, ungrounded review) cannot increase information about the target, whereas an instrument observation introduces a new channel.

10 Discussion and Conclusion

Grounding is the load-bearing variable. Two of the three test-time axes are internal and hit the internal ceiling because they never leave the model’s token space. The third is not

bounded by that ceiling, because a grounded instrument imports a real observation each cycle—it is of course still bounded, by what the instrument can observe and by the proposer’s ability to act on it, which is exactly the coverage principle. We have argued and shown that grounding must hold on *both* sides of the loop. On the feedback side, swapping only the reviewer’s instrument shows improvement tracks coverage: execution feedback converges far cheaper than critique on behavioral bugs, a linter buys nothing on them, and a screenshot-fed VLM reviewer actively regresses slide layout. On the evaluation side, the default VLM-on-a-screenshot metric is structurally blind to the geometric defects the loop fixes, so the entire visual-modality effect is invisible until the metric is grounded in deterministic DOM geometry—and once it is, grounded feedback removes a large, decisive fraction of real layout defects on all four modalities (Figure 10).

A methodological warning for the field. Much of the recent excitement about “LLMs that critique and improve their own visual output” is measured with VLM judges. Our results imply many such measurements are uninformative on layout: the judge cannot see the defects, and using it as the reviewer can make things worse. Where an artifact’s quality is a measurable physical property (geometry, execution, retrieval), the evaluation should be grounded in that property, not in a model’s picture of it.

Grounding is the theory under “loop engineering.” Practitioners have converged on an informal art of *loop engineering*, or *loopcraft*—stacking an inner agent loop with an outer *verification* loop that critiques and revises—as the design pattern for capable agents [15, 20]. That verification loop is exactly the proposer–reviewer loop we study, and our results supply the variable the practitioner framing leaves implicit: *grounding*, qualified by *coverage*, decides when stacking such a loop pays off. A verification loop closed on a model’s opinion—an LLM critique, or a VLM reading a screenshot—is subject to the internal ceiling (Section 2) and, on visual artifacts, actively harmful (Section 5). Only a loop closed on an executed, rendered, or measured observation imports information and clears the ceiling. The lesson is not “add more loops” but “ground the loop you add,” on both the feedback and the evaluation side.

Limits, honestly stated. Two modalities saturate: video editing is already strong single-shot (the lift is real only on a hardened multi-step suite), and deep research saturates because frontier models know well-documented facts and the judge cannot grade facts it does not have. Deterministic geometry applies to artifacts with a measurable intended layout (figures, slides, web, animations); for genuinely flowing content we still rely on a binary rubric, with the VLM-reliability caveat. Our static-form tier is confirmed only on its null prediction (no lift on bugs invisible in form), not yet on a positive one. And because the geometric instrument both supplies the feedback and scores the result, the reported reductions are partly an optimization of the metric itself; we argue in Section 6 (conservative thresholds, non-trivial magnitude, and the ungrounded-reviewer control that moves the same metric the wrong way) that they reflect real, human-visible defects, but a quantitative human-preference validation remains future work.

Future work and broader impact. Two directions follow directly. First, deterministic instruments for modalities beyond layout and execution—differentiable or symbolic checks

for audio, 3D, tabular, and UI-interaction artifacts—would extend grounded evaluation to where model judges are likewise blind. Second, the internal ceiling bounds an information channel; tightening it into a quantitative relationship between an instrument’s coverage and achievable quality is open. On impact: grounding the metric is a cheap, deployable correction to a measurement error that, we argue, currently inflates reported progress on self-improving visual generation. The same instruments are auditable and reduce reliance on opaque model judges. The harness itself is scaffolding over a frozen model, so it inherits that model’s capabilities and failure modes rather than introducing new ones.

Operational recipe. Wrap a frozen frontier model in a proposer–reviewer harness whose instrument covers the defects that matter for the modality. For visual artifacts, ground *both* the reviewer and the scorer in deterministic geometry: measure the rendered DOM, never a screenshot. Allocate the majority of the per-task budget to the proposer, and spend extra compute on samples rather than deeper iteration. For cost-sensitive deployment, distill the proposer into a small student—preserving its sampling variance—and run it inside the same harness. Interaction scaling is real, distinct from reasoning and sampling, and predictable from coverage. Once you ground the metric, it is plainly visible.

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References

- [1] Bradley Brown, Jordan Juravsky, Ryan Ehrlich, Ronald Clark, Quoc V. Le, Christopher Ré, and Azalia Mirhoseini. Large language monkeys: Scaling inference compute with repeated sampling, 2024.
- [2] Stephen Casper, Xander Davies, Claudia Shi, Thomas Krendl Gilbert, et al. Open problems and fundamental limitations of reinforcement learning from human feedback, 2023.
- [3] Thomas M. Cover and Joy A. Thomas. *Elements of Information Theory*. Wiley-Interscience, 2nd edition, 2006.
- [4] DeepSeek-AI. DeepSeek-R1: Incentivizing reasoning capability in LLMs via reinforcement learning, 2025.
- [5] Jonas Gehring, Kunhao Zheng, Jade Copet, Vegard Mella, Taco Cohen, and Gabriel Synnaeve. RLEF: Grounding code LLMs in execution feedback with reinforcement learning. In *International Conference on Machine Learning*, 2025.
- [6] Zhibin Gou, Zhihong Shao, Yeyun Gong, Yelong Shen, Yujiu Yang, Nan Duan, and Weizhu Chen. CRITIC: Large language models can self-correct with tool-interactive critiquing. In *International Conference on Learning Representations*, 2024.

- [7] Dingkun Hong. The practice and exploration of AI coding. Keynote, ByteDance/Volcano Engine Force Conference, Beijing; speaker is VP of Engineering at ByteDance, 2026. <https://mp.weixin.qq.com/s/tJAinVKzZAqZrhiEvGU2Dg>.
- [8] Jie Huang, Xinyun Chen, Swaroop Mishra, Huaixiu Steven Zheng, Adams Wei Yu, Xinying Song, and Denny Zhou. Large language models cannot self-correct reasoning yet. In *International Conference on Learning Representations*, 2024.
- [9] Ryo Kamoi, Yusen Zhang, Nan Zhang, Jiawei Han, and Rui Zhang. When can LLMs actually correct their own mistakes? a critical survey of self-correction of LLMs. *Transactions of the Association for Computational Linguistics*, 2025.
- [10] Zimu Lu, Houxing Ren, Yunqiao Yang, Ke Wang, Zhuofan Zong, Junting Pan, Mingjie Zhan, and Hongsheng Li. WebGen-Agent: Enhancing interactive website generation with multi-level feedback and step-level reinforcement learning, 2025.
- [11] Aman Madaan, Niket Tandon, Prakhar Gupta, Skyler Hallinan, Luyu Gao, Sarah Wiegrefe, Uri Alon, Nouha Dziri, Shrimai Prabhumoye, Yiming Yang, Sean Welleck, Bodhisattwa Prasad Majumder, Shashank Gupta, Amir Yazdanbakhsh, and Peter Clark. Self-refine: Iterative refinement with self-feedback. In *Advances in Neural Information Processing Systems*, 2023.
- [12] Shishir G. Patil, Tianjun Zhang, Xin Wang, and Joseph E. Gonzalez. Gorilla: Large language model connected with massive APIs. In *Advances in Neural Information Processing Systems*, 2024.
- [13] Pine AI. Pine AI: The most natural human-computer interface is your voice. Blog post, 2026. URL <https://www.19pine.ai/blog/pine-ai-the-most-natural-human-computer-interface-is-your-voice>. Accessed 2026-06-28.
- [14] Yangjun Ruan, Eleftheria Briakou, Charles Xie Han, Yu Chen, Yufan Jiao, et al. Iterative agent decoding for detecting compounding errors in LLM agents, 2025.
- [15] Sydney Runkle. The art of loop engineering. LangChain (blog), 2026. <https://www.langchain.com/blog/the-art-of-loop-engineering>.
- [16] Timo Schick, Jane Dwivedi-Yu, Roberto Dessí, Roberta Raileanu, Maria Lomeli, Luke Zettlemoyer, Nicola Cancedda, and Thomas Scialom. Toolformer: Language models can teach themselves to use tools. In *Advances in Neural Information Processing Systems*, 2023.
- [17] Junhong Shen, Hadi Jain, Zelai Xiao, Brandon Amos, Aaditya Ramdas, Yuandong Tian, and Alborz Geramifard. Thinking vs. doing: Agents that reason by scaling test-time interaction. In *Advances in Neural Information Processing Systems*, 2025.
- [18] Noah Shinn, Federico Cassano, Edward Berman, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. Reflexion: Language agents with verbal reinforcement learning. In *Advances in Neural Information Processing Systems*, 2023.

- [19] Charlie Snell, Jaehoon Lee, Kelvin Xu, and Aviral Kumar. Scaling LLM test-time compute optimally can be more effective than scaling model parameters. In *International Conference on Learning Representations*, 2025.
- [20] swyx and Latent Space. Loopcraft: The art of stacking loops. Latent Space (blog), 2026. <https://www.latent.space/p/ainews-loopcraft-the-art-of-stacking>.
- [21] Kimi Team. Kimi k1.5: Scaling reinforcement learning with LLMs, 2025.
- [22] Dat Tran and Douwe Kiela. Single-agent LLMs outperform multi-agent systems on multi-hop reasoning under equal thinking token budgets, 2026.
- [23] Guanzhi Wang, Yuqi Xie, Yunfan Jiang, Ajay Mandlekar, Chaowei Xiao, Yuke Zhu, Linxi Fan, and Anima Anandkumar. Voyager: An open-ended embodied agent with large language models. In *NeurIPS Foundation Models for Decision Making Workshop*, 2023.
- [24] Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc V. Le, Ed H. Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. Self-consistency improves chain of thought reasoning in language models. In *International Conference on Learning Representations*, 2023.
- [25] Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed H. Chi, Quoc V. Le, and Denny Zhou. Chain-of-thought prompting elicits reasoning in large language models. In *Advances in Neural Information Processing Systems*, 2022.
- [26] Shunyu Yao, Dian Yu, Jeffrey Zhao, Izhak Shafran, Thomas L. Griffiths, Yuan Cao, and Karthik Narasimhan. Tree of thoughts: Deliberate problem solving with large language models. In *Advances in Neural Information Processing Systems*, 2023.
- [27] Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. ReAct: Synergizing reasoning and acting in language models. In *International Conference on Learning Representations*, 2023.
- [28] Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. Judging LLM-as-a-judge with MT-bench and chatbot arena. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

A Formal Statement of the Internal Ceiling

This appendix formalizes the one-sentence argument of Section 2.3. Let A^* be the (task-determined) correct artifact and A a candidate. The proposer is a channel from its inputs Θ (weights) and prompt X to A . A *reviewer* that only re-reads A and reasons—ungrounded critique—forms the Markov chain $A^* \rightarrow (\Theta, X) \rightarrow A \rightarrow R$, where R is its critique. By the data-processing inequality [3],

$$I(A^*; R) \leq I(A^*; (\Theta, X)),$$

so no amount of reasoning or ungrounded review can recover more information about A^* than is already in (Θ, X) . This bounds reasoning-only scaling. Sampling is bounded differently

but no less tightly: for any finite N , best-of- N can only *select* among the N candidates the proposer actually draws, so even an oracle (grounded) verifier cannot return a correct artifact the proposer is too unlikely to sample within the budget. Its practical ceiling is the high-probability support of the proposer’s own output distribution, not a new information channel.

Grounded feedback escapes both ceilings by feeding an instrument observation $E = g(A, \text{world})$ —a test outcome, a measured bounding box, a search verdict—back into *generation*. Conditioning the next proposal on E lets the proposer synthesize a candidate it would not otherwise have sampled: formally, $I(A^*; (R, E))$ can exceed $I(A^*; (\Theta, X))$ because E carries information obtained by running the artifact against reality (the tests encode the specification; the layout engine encodes real rendering semantics the model only approximates). The *coverage* qualification of Section 2.2 enters here: E helps only to the extent that g ’s observation is informative about the defect classes actually present—a linter’s g observes form, so on runtime-logic bugs $I(A^*; E)$ adds nothing actionable, which is what Figure 6 measures.

The symmetric claim covers the scorer. A metric M that is itself a model’s function of a lossy view of A (a VLM judging a cropped screenshot) cannot certify an improvement that lives in the part of A its channel drops. Grounding the evaluation is therefore a precondition for *detecting* interaction scaling on any modality whose quality is not fully visible to the default judge.

Two cautions. The inequality bounds the *information channel*, not achievable quality directly; we use it to motivate the saturation that Figure 3 then measures, and treat the measurement as the evidence. And grounded interaction is of course still bounded—by what the instrument exposes and by the proposer’s ability to act on it.

B Detailed Results and Configurations

This appendix collects the full tables underlying the figures in the main text, from the four-strategy scaling curves through the distillation ablations to the original six-modality holistic overview (the last superseded for the visual modalities by the grounded measurements of Section 6).

B.1 Four-strategy scaling at a matched 20K-token budget

Table 2. The headline scaling result (3-seed mean $\pm$ SD), underlying Figure 3. Pass rate by strategy $\times$ budget on 15 hard code tasks, Sonnet 4. S, L, and H are averaged over three independent on-policy runs. R was run once (extended-thinking variation across seeds is small at temp 0) at the sweep’s budget partition (a more generous partition lifts R to 86.7%, see Table 3). S’s small dip from $B=1K$ (80.0%) to $B=5K$ (77.8%) is within the ± 3.1 pp seed noise on a 3-seed $\times$ 15-task suite, since pass@ N is non-decreasing in N in expectation but not pointwise across finite-sample on-policy seeds. *Reasoning-only and best-of- N saturate well below 100%*, while all three iteration-with-feedback strategies reach $\geq 97.8\%$ at $B=20K$ (L hit 100% on 2 of 3 seeds, H on all 3, so H is the only variant at strict 100% across seeds). The architectural ordering across H, L is within seed noise on pass rate; what separates them is token efficiency and reliability (Figure 5).

Budget B	R (reasoning-only)	S (best-of- N)	L (single-agent loop)	H (proposer-reviewer)
1K	60.0%	80.0% ± 0.0 pp	57.8% ± 3.1 pp	62.2% ± 3.1 pp
5K	73.3%	77.8% ± 3.1 pp	93.3% ± 0.0 pp	91.1% ± 3.1 pp
20K	73.3%	86.7% ± 0.0 pp	97.8% ± 3.1 pp	100.0% ± 0.0 pp
Ceiling	73.3%	86.7%	$\sim 100\%$	100% (zero variance)

B.2 Reasoning-only at a matched budget

Table 3. Reasoning-only at matched budget vs. single-shot and the harness on the 15 hard code tasks. Single-seed numbers (the Phase-1 reference seed, also used in Table 4). The 3-run aggregate is reported in Table 17 ($66.7 \pm 6.7\%$ single-shot, $100.0 \pm 0.0\%$ reviewed). *Reading the numbers:* single-shot here is this single seed’s $11/15 = 73.3\%$ (the canonical 3-seed mean is 66.7%), and the reasoning value shown is the *thinking-heavy* partition (86.7%). This coincides numerically with the oracle best-of- N ceiling, while the *default*-partition reasoning ceiling (73.3%, Table 2) coincides with this seed’s single-shot. Both are coincidences of the small suite, not the same run. Reasoning closes two-thirds of the harness’s gap over single-shot but falls 6.7 pp short of the harness’s pass rate even at $1.57\times$ the harness’s token budget.

Strategy	Pass rate	Mean tokens	Δ vs. single-shot
Single-shot	73.3% (11/15)	1,406	—
Reasoning-only (thinking-heavy partition)	86.7% (13/15)	4,137	+13.3 pp
Proposer-reviewer harness	93.3% (14/15)	2,643	+20.0 pp

B.3 Feedback-channel control on a fixed code suite

Table 4. Feedback-channel controls on 15 hard code tasks (Sonnet 4 proposer and reviewer), underlying Figure 6. Each row is a single on-policy seed. The “SS” column is the *per-seed* single-shot pass rate (not held constant across rows) and “Reviewed” is the ceiling after that seed’s harness run. The three SS values (10/15 and 11/15) are draws from the same single-shot distribution that gives Table 17’s 3-run mean of $66.7 \pm 6.7\%$ (i.e. within the ± 1 SD band centered on 10/15). Because the SS column varies by row, the load-bearing comparison is the *reviewed* ceiling: execution reaches 14/15, critique and critique+linter reach 13/15 (+6.7 pp), and execution is $\sim 2.5\times$ more token-efficient (2,643 vs. $\sim 7,100$ tokens/task, 1.53 vs. 2.07 iterations). The single task unrecovered in every control row is code_011 (CJK text wrapping). The full harness at $B=20K$ recovers it across all three seeds (Table 2), which is why the headline reviewed rate is the strict 100% of Table 17 while this leaner single-seed control caps at 14/15.

Feedback channel	Reviewer sees	SS	Reviewed	Iters
None (single-shot)	—	11/15	—	—
Ungrounded critique	code only	10/15	13/15	2.07
+ linter (static form)	code + ruff output	11/15	13/15	2.07
+ execution	code + test stderr/traceback	11/15	14/15	1.53

B.4 Single-agent loop vs. proposer–reviewer harness

Table 5. Three interaction-strategy variants at $B=20K$ on the 15 hard code tasks, all using execution feedback; underlying Figure 5. Pass rate is within seed noise (sign-test $p>0.6$ on the H–L comparison). The harness wins on *token efficiency* (1,029 vs. 1,431 L vs. 1,416 IAD mean output tokens) and on *seed variance* (zero across three seeds for H, while L hit 100% in 2 of 3 seeds and IAD was run for one seed only).

Strategy	Pass rate	Mean tokens	Notes
L (single-agent loop)	97.8% ± 3.1 pp (3 seeds)	1,431	1 task in 1 seed
IAD (oracle, $K=3$, $T=0.7$)	100.0% (1 seed)	1,416	+38% tokens vs. H
H (proposer–reviewer)	100.0% ± 0.0 pp (3 seeds)	1,029	zero seed variance

B.5 Deterministic geometric-feedback statistics

Table 6. Full statistics for Figure 10: deterministic geometric-feedback results, one identical configuration across four visual modalities (Sonnet 4, $T=0$, 3 seeds, alignment-inclusive reward). SS/Rev. are mean defects per artifact, single-shot vs. reviewed; Δ is the mean defect reduction; CI is a paired bootstrap 95% interval on $\Delta\%$ (10k resamples); p is a two-sided paired sign test over decisive (task, seed) pairs. Web is summed over 1920/375 widths, animations over sampled frames.

Modality	n	SS	Rev.	Δ	95% CI	impr./regr. (p)
Academic figures (20)	60	0.57	0.15	−74%	[52, 89]%	17/2 (7×10^{-4})
Dense slides (12)	36	1.25	0.33	−73%	[45, 93]%	13/1 (1.8×10^{-3})
Web pages (20)	60	16.1	8.5	−47%	[30, 62]%	33/2 (4×10^{-8})
Animations (20)	60	16.9	10.2	−40%	[10, 64]%	34/7 (3×10^{-5})

B.6 Cross-model replication (code)

Table 7. Cross-model replication on the same 15 hard code tasks, each family over *three* independent on-policy seeds at $T=0.7$ (Claude is the multi-run reference from Table 17); underlying Figure 11a. The reviewed ceiling has zero seed variance for all three families, and the harness lift survives across Anthropic, Alibaba, and OpenAI. Single-shot per-seed pass rates: Qwen 66.7/73.3/73.3%, GPT-5 86.7/80.0/73.3%.

Model	Single-shot pass	Reviewed pass	Δ (lift)
Claude Sonnet 4 (3 seeds)	$66.7 \pm 6.7\%$	$100.0 \pm 0.0\%$	+33.3 pp
Qwen3-235B-Instruct-2507 (3 seeds)	$71.1 \pm 3.8\%$	$93.3 \pm 0.0\%$	+22.2 pp
GPT-5 (3 seeds)	$80.0 \pm 6.7\%$	$100.0 \pm 0.0\%$	+20.0 pp

B.7 Held-out generalization (code)

Table 8. Held-out generalization on 32 newly constructed code tasks (zero overlap with the 15-task development set). The harness recovers 3/3 single-shot failures and regresses zero passing tasks. The smaller absolute Δ is a baseline-compression effect (single-shot is 90.6% on the held-out set, leaving at most 9.4 pp headroom).

Split	N	Single-shot	Reviewed	Δ	SS-fail fixes / regr.
Development (3-run mean)	15	$66.7 \pm 6.7\%$	$100.0 \pm 0.0\%$	+33.3 pp	5/5 / 0
Held-out v2	32	90.6%	100.0%	+9.4 pp	3/3 / 0

B.8 Budget-allocation simplex

Table 9. Budget allocation simplex on 15 hard code tasks, $B = 10K$ total; underlying Figure 12. Pass rate is monotone in the proposer’s per-call cap, and review-heavy corners (low b_1) collapse. The 9-point sweep yields an 86.6 pp spread.

Allocation	b_1 (prop)	b_2 (exec)	b_3 (rev)	Pass rate	Mean tokens
A (propose-heavy)	0.80	0.10	0.10	93.3% (14/15)	3,101
G (prop-dominant)	0.50	0.25	0.25	93.3% (14/15)	2,697
D (prop+exec)	0.40	0.40	0.20	80.0% (12/15)	4,025
E (prop+review)	0.40	0.20	0.40	80.0% (12/15)	4,388
I (equal)	0.33	0.34	0.33	66.7% (10/15)	4,845
H (review-dominant)	0.25	0.25	0.50	40.0% (6/15)	7,485
F (exec+review)	0.20	0.40	0.40	13.3% (2/15)	8,771
B (execute-heavy)	0.10	0.80	0.10	6.7% (1/15)	9,383
C (review-heavy)	0.10	0.10	0.80	6.7% (1/15)	9,383
Spread:				86.6 pp	

B.9 Token ROI by modality

Table 10. Per-task token cost and quality return-on-investment for the harness, averaged across three on-policy runs. “Extra tokens” is reviewed-minus-single-shot per task. Tokens per 0.01 quality is extra tokens divided by (mean reviewed quality – mean single-shot quality)·100. Code dominates by 100×.

Modality	Single-shot tokens	Reviewed tokens	Extra tokens	Tokens / 0.01 quality
Code	3,336	4,575	1,239	37
Video	3,808	20,584	16,776	359
Research	14,116	27,793	13,677	2,415
Webpages	14,060	80,148	66,088	5,128
Animations	25,491	89,009	63,518	4,331
Slides	10,274	36,956	26,682	4,447

B.10 Distillation: headline

Table 11. Distilled 8B student on the 18-task hard held-out suite (teacher’s pre-curated single-shot failures, so teacher single-shot is 0/18 by construction). The student lifts judge-keep from 0% to 44% at pass@1 and 56% at pass@3. The $0.51 \times$ mean@1 capability ratio against the teacher is established on a separate 44-task out-of-distribution suite (Table 12), where teacher single-shot is non-degenerate.

Setup	Compute	Judge-keep
1 sample $\times$ 5 turns	1×	44% (8/18)
1 sample $\times$ 8 turns	1.6×	28% (regressed)
3 samples $\times$ 5 turns (pass@3)	3×	56% (10/18)

B.11 Distillation: out-of-distribution capability ratio

Table 12. Distilled 8B student (V3 + force-close) on a 44-task out-of-distribution suite where the teacher’s single-shot is non-degenerate (teacher 20/44). Capability ratio is the student’s judge-keep divided by the teacher’s 20/44. Two sampling seeds are used, and pass@2 is the union over both. This is the suite that establishes the $\sim 0.51 \times$ mean@1 / $0.70 \times$ pass@2 ratios quoted in Section 8 and Figure 13.

Metric	sample 1	sample 2	mean@1	pass@2
Judge-keep	9 / 44 (20%)	11 / 44 (25%)	23%	32%
Capability ratio (vs teacher 20/44)	0.45×	0.55×	0.51×	0.70×

B.12 Distillation: capability U-curve over force-close budget

Table 13. Reasoning-cap U-curve on the 18-task hard held-out suite. SFT recipe sweep on Qwen3-VL-8B-Instruct (37 judge-kept teacher traces, same inference protocol). The optimum is at the student’s coherence horizon, not the teacher’s.

Variant	Reasoning cap (chars)	Judge-keep	Identical-retry
V1 (no reasoning)	0	6%	67%
V4	1000	11%	35%
V2	3000	17%	32%
V3 (chosen)	1500	44%	10%

B.13 Distillation: data-scale

Table 14. Three independent attempts to scale the SFT corpus past V3’s 37 examples. All regressed. Quality dominates quantity at this scale.

Variant	Composition	n examples	Judge-keep
V3 (chosen)	235B teacher, judge-kept	37	44%
V5	V3 + 13 judge-rejected (235B)	50	33%
V6	V3 + 20 judge-kept (30B teacher)	57	33%
RFT v1	Self-rolled, judge-filtered	61	39%

B.14 Distillation: RFT regresses pass@3

Table 15. RFT polishes consistency at the cost of pass@ k ; underlying Figure 14. Per-turn metrics improve, per-task variance collapses, and pass@3 regresses by 17 pp. For deployment with sampling, the model’s variance *is* the inference-scaling budget. *Note on notation.* **mean@1** is the expected pass rate of a single random sample, computed as the mean accuracy across the three sampling seeds (one $T=0$ greedy + two $T=0.7$). This is the natural variance-sensitive metric and differs from the greedy single-seed $pass@1=44\%$ reported in Table 11, which is just seed 1’s accuracy. $pass@2$ and $pass@3$ are union pass rates over the first 2 / all 3 seeds.

	Consistency metrics			Sampling-aware judge-keep		
	id-retry ↓	addresses-rev ↑	no-artifact ↓	mean@1	pass@2	pass@3
V3 (chosen)	9%	83%	2	31%	50%	56%
RFT v1	8%	89%	1	28%	39%	39%

B.15 Per-task pass@ k for the 8B markdown student**Table 16.** Per-task judge-keep outcome for the 8B markdown student on the 18-task hard held-out suite.
✓ = judge-kept, · = judge-rejected.

Category	Task	seed 1 ($T=0$)	seed 2 ($T=0.7$)	seed 3 ($T=0.7$)	pass@3
webpages	web_002	✓	·	·	✓
	web_003	✓	·	✓	✓
	web_005	·	·	·	·
	web_008	·	·	·	·
	web_012	·	·	·	·
slides	slide_001	·	·	·	·
	slide_006	·	·	·	·
	slide_007	·	·	·	·
	slide_008	✓	·	·	✓
	slide_010	·	·	·	·
	slide_014	✓	✓	✓	✓
	slide_018	✓	✓	·	✓
	slide_020	·	✓	·	✓
code	code_h001	✓	·	·	✓
	code_h002	✓	✓	·	✓
	code_h003	·	·	·	·
	code_h004	·	·	✓	✓
	code_h005	✓	✓	✓	✓
Total kept:		8/18	5/18	4/18	10/18 (56%)

B.16 Original six-modality holistic overview (historical)

Table 17. Single-shot vs. reviewed quality across six modalities, Claude Sonnet 4. Means and standard deviations are computed across three independent on-policy runs (per-run means, then aggregated). Code is binary pass rate, and other modalities are continuous in $[0, 1]$. p -values: one-sided paired sign test on per-task means. **Bold** marks rows where the lift clears $p < 0.05$. Code is scored by execution (pytest), video by a native full-video rubric (Gemini 3.1 Pro judging the whole rendered clip), and the four visual/factual rows use a holistic screenshot judge and serve only as a cross-modal overview. *This table is historical and is superseded for the visual modalities by the grounded measurements of Section 6: under the deterministic DOM-geometry instrument (Table 6) the figure, slide, web, and animation effects are large and significant, whereas the holistic-judge numbers here cannot see the geometric defects and therefore mis-state them. The visual rows here also use the earlier, smaller task sets (15 web and 15 animation tasks, 20 generic slides), not the hardened suites of Table 6 (20/20/12 real-paper), which is why the per-modality N differs between the two tables. The deep-research and video rows also use different instruments than Section 6. Research here is the old holistic judge (0.63 $\rightarrow$ 0.69) rather than the search-grounded factual rubric (single-shot 0.982), so the large numeric gap is an instrument change, not a contradiction. Video uses the same Gemini-native rubric in both places (0.70 $\rightarrow$ 0.74).*

Modality	N	Feedback channel	Single-shot	Reviewed	Δ (sign-test p)
Code	15	execution	$66.7 \pm 6.7\%$	$100.0 \pm 0.0\%$	+33.3 pp (0.008)
Video editing	15	exec.+temporal	0.70 ± 0.03	0.74 ± 0.05	+0.04 (0.64)
Deep research	15	external facts	0.63 ± 0.06	0.69 ± 0.06	+0.06 (0.090)
Web pages	15	geometry	0.43 ± 0.03	0.56 ± 0.01	+0.13 (0.073)
Animations	15	temporal	0.40 ± 0.03	0.55 ± 0.01	+0.15 (0.073)
Slide decks	20	geometry	0.73 ± 0.01	0.79 ± 0.01	+0.06 (0.090)